

FARM MANAGEMENT SYSTEM

User Stories

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, harvest, and lower information so that all operational data is stored securely and systematically. I want to enter and update farm details, crop production records, and lower transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1.INTRODUCTION

1.1 Project Overview:

"The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions."

1.2 Purpose:

The main purpose of this project is to:

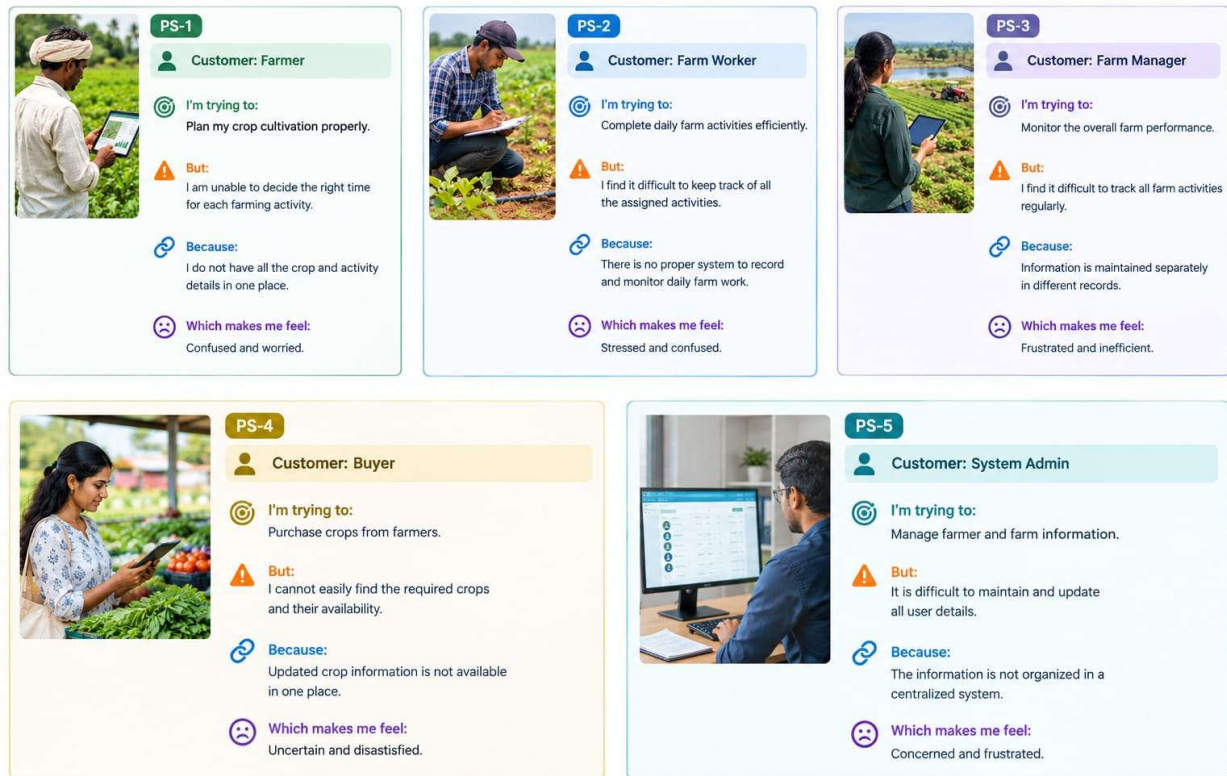
- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Plan my crop cultivation properly	I am unable to decide the right time for each farming activity	I do not have all the crop and activity details in one place	Confused and Worried
PS-2	Farm Worker	Complete	I am unable	Tasks are	Stressed

		my daily field work on time	to track my assigned tasks properly	given verbally and there is no system to record daily activities	and unorganized
PS-3	Farm Manager	Monitor the overall farm performance	I find it difficult to track all farm activities regularly	Information is maintained separately in different records	Frustrated and inefficient
PS-4	Buyer	Purchase crops from farmers directly	I cannot easily find the required crops and their availability	There is no centralized system to show live crop stock	Uncertain and dissatisfied
PS-5	System Admin	Manage farmer and farm information	It is difficult to maintain and update all user details	The information is not organized in a centralized system	Concerned and frustrated



2.2 Empathy Map Canvas

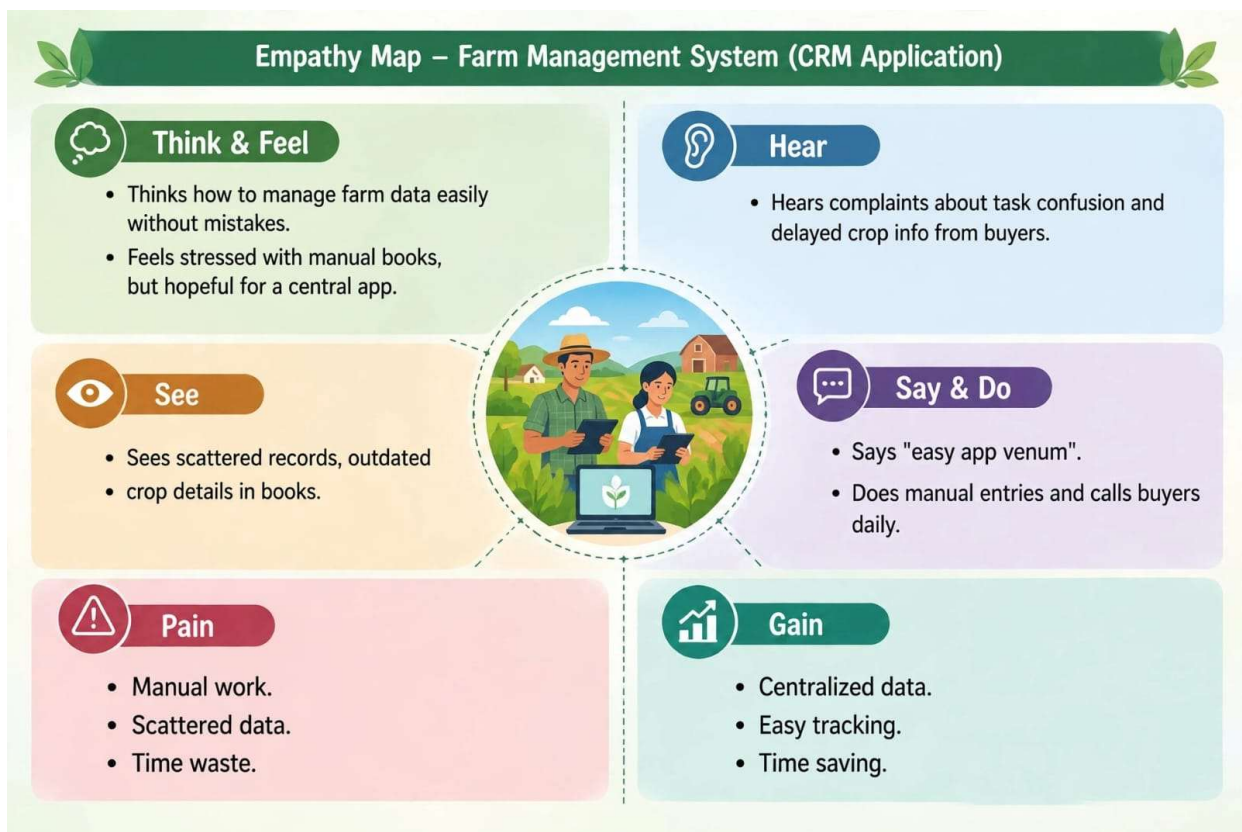
Target users:

- Farmers
- Farm Workers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis:

- **Think & Feel:** Thinks how to manage farm data easily without mistakes. Feels stressed with manual books, but hopeful for a central app.
- **Hear :** Hears complaints about task confusion and delayed crop info from buyers.

- **See** : Sees scattered records,outdated crop details in books.
- **Say & Go** : Says “easy app venum”.Does manual entires and calls buyers easily.
- **Pain** : Manual work,scattered data,time waste.
- **Gain** : Centralized data, easy tracking,time saving.



2.3 Brainstorming

Brainstormed ideas

- Centralized Farmer Database System
- Smart Crop Activity & Task Assignment

- Live Crop Availability Dashboard for Buyers
- Farm Expense & Profit Tracker
- Role-Based Salesforce Farm Management System

Selected idea

Role-Based Salesforce Farm Management System



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages :

1. Secure Role-Based Login
2. Centralized Dashboard Access
3. Farm & Crop Data Entry
4. Farmer & Buyer Management
5. Task Assignment for Workers
6. Expense & Profit Tracker
7. Reports for Decision Making



3.2 Solution Requirements

Functional Requirements

FR	Requirement
FR-1	User Login & Authentication
FR-2	Dashboard Management
FR-3	Farm & Crop Data Management
FR-4	Farmer & Buyer Management
FR-5	Task Assignment & Tracking
FR-6	Expense Tracker
FR-7	Reports & Analytics

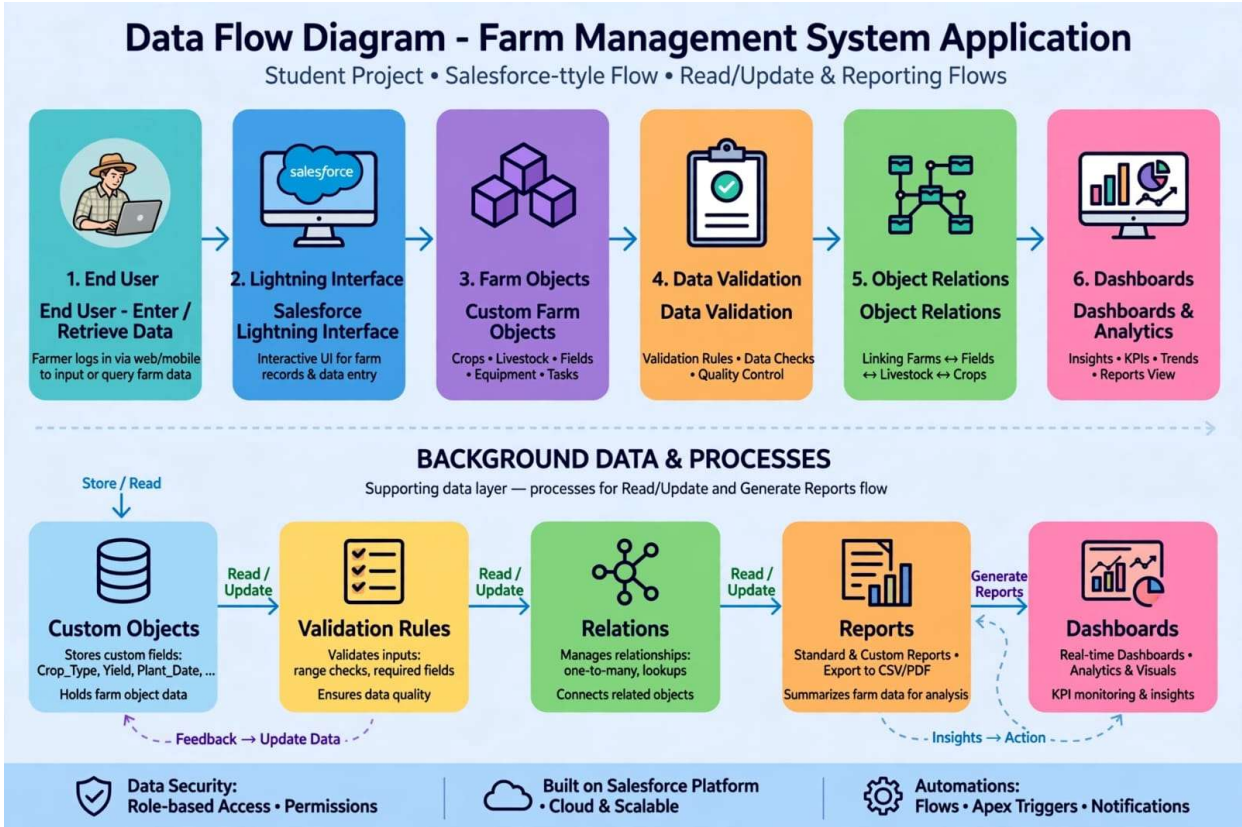
Non-Functional Requirements

NFR	Description	Example
Usability	System should have simple and user-friendly for farmers with less technical knowledge	Easy navigation with icons and regional language support
security	System should protect farm,crop and buyer data farm unauthorized access	Role-based access-only Admin can delete data
Reliability	System should store all data correctly without any loss	Auto backup even if internet fails
Performance	System should load dashboard and reports quickly	Reports should load within 3 seconds
Availability	System should be available anytime for farmers	24/7 access through cloud platform
Scalability	System should handle more farms,users and data in future	Supports 100+ farmers without slow performance

3.3 Data Flow Diagram

Data Flow

End User -> Authentication -> Lightning UI -> Custom -> Farm Objects -> Validation Checks -> Inter-Objects Relationships -> Reports -> Dashboard Visualization



3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI with Custom Components

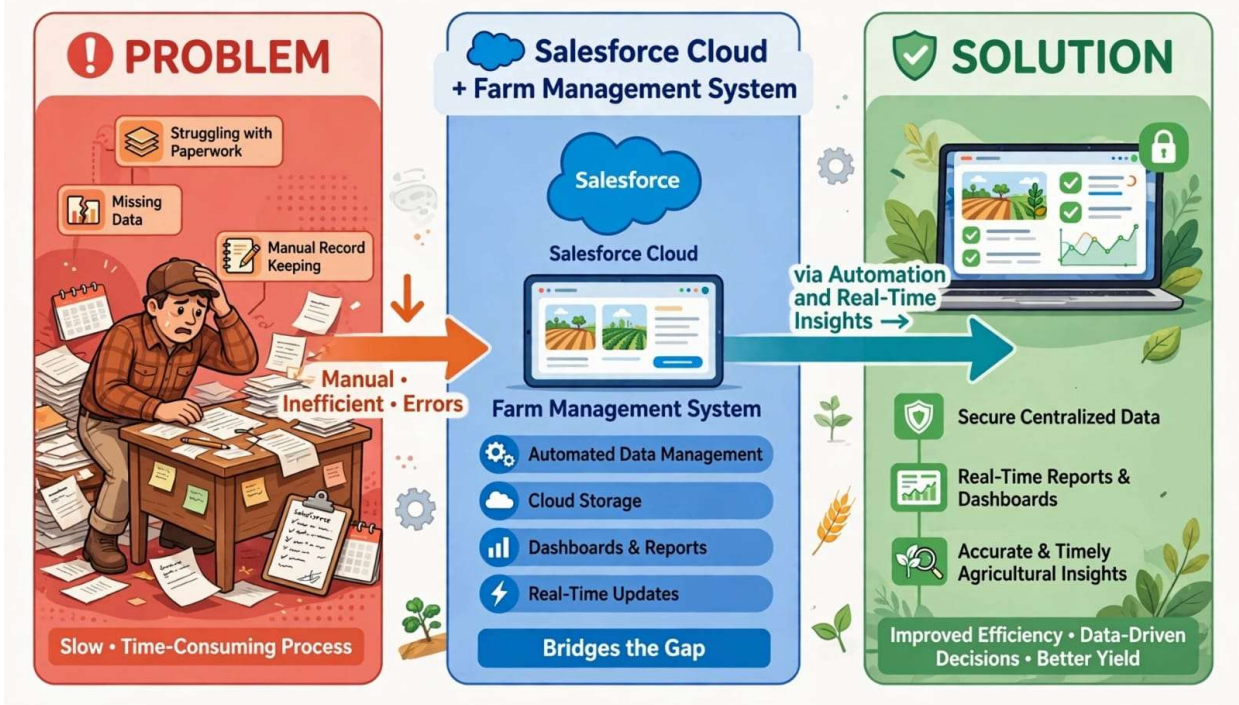
Backend	Salesforce Platform (Apex Classes,Triggers)
Database	Salesforce cloud storage with Custom objects
Automation	Validation Rules,Flows and Process Automation
Analytics	Dynamic Reports & Interactive Dashboards

4.PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

Farmers currently depend on manual notes and scattered Excel sheets for crop tracking, which causes missing data and delayed decisions. Our Farm Management System built on Salesforce converts this manual work into a centralized cloud system. With automated workflows,secure data storage,and live dashboards,farmers get accurate crop and buyer details instantly without data loss or time delay .

Problem-Solution Fit - Farm Management System



4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Farmers currently store farm,crop,farmer and buyer details in notebooks and Excel sheets,which creates errors,missing data and reporting delays.
Solution	Developing a centralized Farm Management System on Salesforce Lightning Platform with automated data handling and secure cloud database.
Innovation	Includes smart data validation,

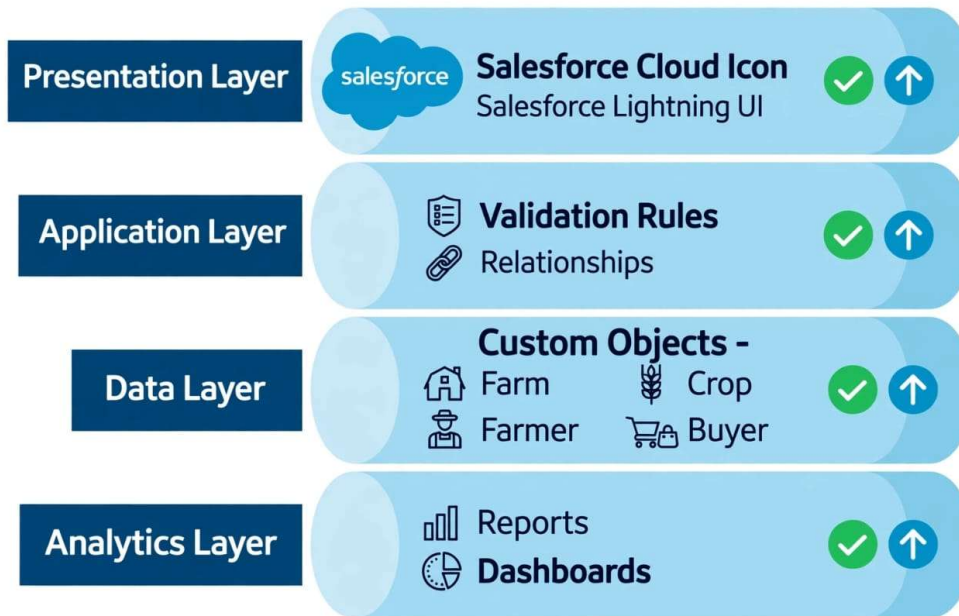
	auto-linking between Farm-Crop-Buyer objects,real-time cloud sync and instant dashboards analytics.
Impact	Increases data accuracy to 99%,reduces manual work time,provides faster crop reports and helps farmers take better decisions.
Business Model	A digital agriculture service platform offering farm data management ,crop tracking and buyer connection services.
Scalability	Designed to support unlimited farms, users and locations through Salesforce scalable cloud infrastructure.

4.3 Solution Architecture

Architecture Layers

1. **Presentation** - Interactive Salesforce Lightning Interface.
2. **Application Layer** - Business Logic with Validation Rules,Flows and Relationships.
3. **Data Layer** - Custom Objects Storage.
4. **Analytics Layer** - Smart Reports & Dashboards.

Solution Architecture - Farm Management System



Student Project • Modular • Scalable • Secure

5.PROJECT PLANING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards

- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule.

Sprint	Functional Requirement (Epic)	User Story Number	User Story/Task	Story points	Priority	Team Member
Sprint-1	Developer setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member2
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationship	5	High	Member2

			s so that agricultural data is stored and managed efficiently			
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member1
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I	3	High	Member4

			want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention .			
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction Data is secure.	5	High	Member1
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I	4	High	Member5

			can monitor farm performance and Make informed decisions.			
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity , sales trends, and operational statistics so that I can monitor performance visually.	4	High	Member3

Project Tracker,Velocity & Burndown Chart :

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date(Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	2 Days	23 Sep 2026	24 Sep 2026	3	24 Sep 2026
Sprint-2	20	1 Days	25 Sep 2026	25 Sep 2026	5	25 Sep 2026

Sprint-3	20	2 Days	26 Sep 2026	27 Sep 2026	3	27 Sep 2026
Sprint-4	20	1 Days	28 Sep 2026	28 Sep 2026	4	28 Sep 2026

6.PROJECT DEVELOPMENT PHASE :

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What is Salesforce?

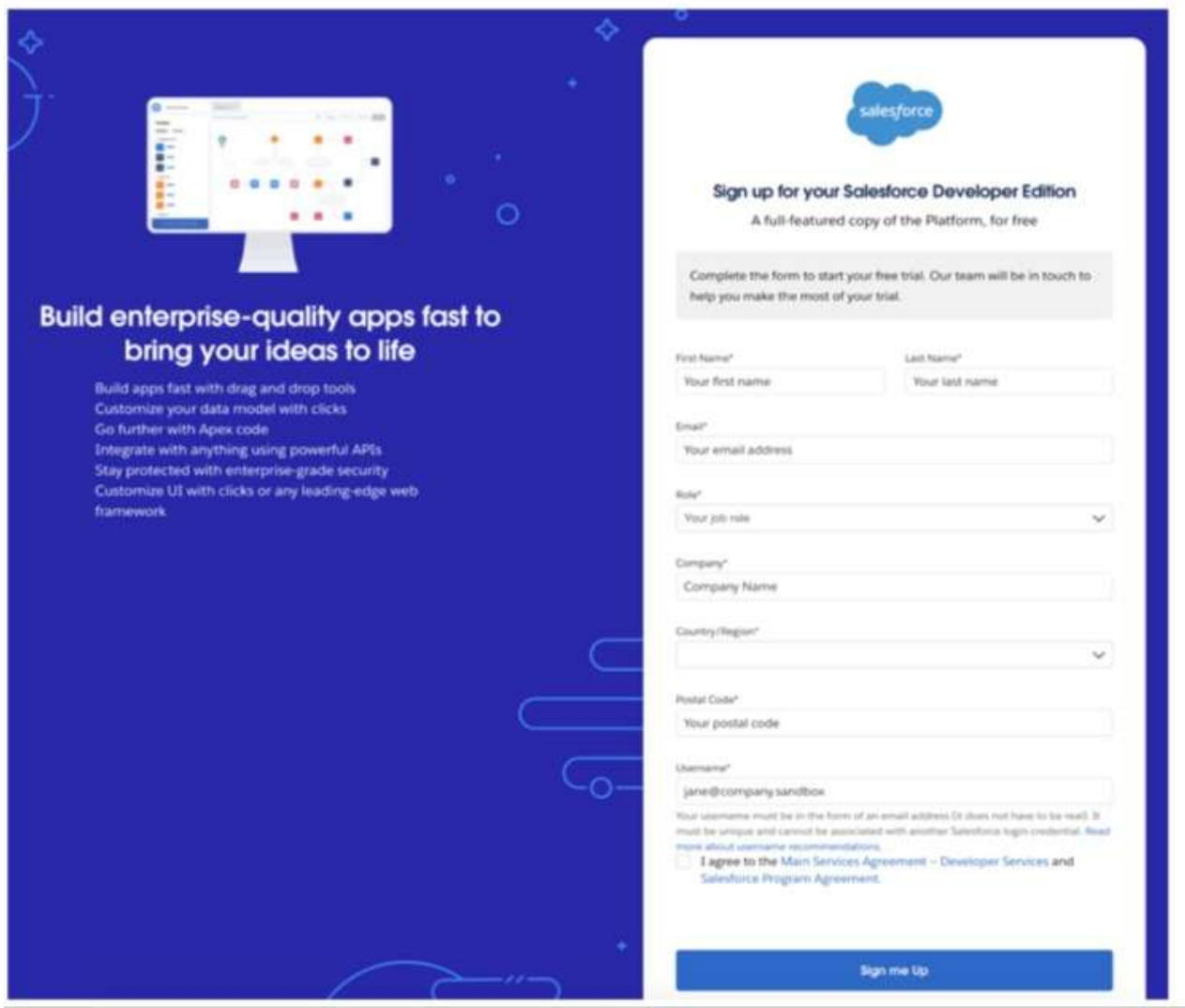
Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup) .
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



The image shows the Salesforce Developer Edition sign-up page. On the left, a blue background features a monitor icon displaying a Salesforce interface and a list of capabilities: 'Build apps fast with drag and drop tools', 'Customize your data model with clicks', 'Go further with Apex code', 'Integrate with anything using powerful APIs', 'Stay protected with enterprise-grade security', and 'Customize UI with clicks or any leading-edge web framework'. On the right, a white sign-up form is set against a blue background. It includes the Salesforce logo, the title 'Sign up for your Salesforce Developer Edition', and the subtitle 'A full-featured copy of the Platform, for free'. A grey box contains the text: 'Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.' The form fields are: 'First Name*' (with placeholder 'Your first name'), 'Last Name*' (with placeholder 'Your last name'), 'Email*' (with placeholder 'Your email address'), 'Role*' (a dropdown menu with placeholder 'Your job role'), 'Company*' (with placeholder 'Company Name'), 'Country/Region*' (a dropdown menu), 'Postal Code*' (with placeholder 'Your postal code'), and 'Username*' (with placeholder 'jane@company.sandbox'). Below the username field, a note states: 'Your username must be in the form of an email address. (It does not have to be real). It must be unique and cannot be associated with another Salesforce login credential. Read more about username recommendations.' A checkbox is followed by the text 'I agree to the Main Services Agreement – Developer Services and Salesforce Program Agreement.' At the bottom is a blue 'Sign me Up' button.

Build enterprise-quality apps fast to bring your ideas to life

- Build apps fast with drag and drop tools
- Customize your data model with clicks
- Go further with Apex code
- Integrate with anything using powerful APIs
- Stay protected with enterprise-grade security
- Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*
Your first name

Last Name*
Your last name

Email*
Your email address

Role*
Your job role

Company*
Company Name

Country/Region*
Country/Region

Postal Code*
Your postal code

Username*
jane@company.sandbox

Your username must be in the form of an email address. (It does not have to be real). It must be unique and cannot be associated with another Salesforce login credential. Read more about username recommendations.

☐ I agree to the Main Services Agreement – Developer Services and Salesforce Program Agreement.

Sign me Up

6..By using username and password you can into the Salesforce org.The homepage will appear below.

Setup

Home

Object Manager

Quick Find

Setup Home

Salesforce Go

Commerce Setup Assistant

Hyperforce Assistant

Release Updates

Salesforce Mobile App

Lightning Usage

AgentExchange Offers

Sales Cloud Everywhere

ADMINISTRATION

Users

Data

Email

PLATFORM TOOLS

Subscription Management

Apps

Feature Settings

Slack

Data Cloud

Heroku

Search Setup

Welcome, Deepika

Manage and customize Salesforce from Setup. Browse suggestions, explore features, and more.

Achieve Popular Business Goals

Cross Cloud

Connect with Sales Prospects and Customers

Help sales reps prioritize and engage with customers.

Includes

Sales Cloud Everywhere

Agentforce (Default)

Einstein Sales Emails

3 Completed

1 In Progress

Keep Going

Cross Cloud

Track & Manage Customer Data

Collect, organize, and analyze customer data to drive insights, personalize experiences, and boost engagement.

Includes

Case Management

Feedback Management

Einstein Activity Capture

2 Completed

1 In Progress

Keep Going

Field Service

Schedule Field Service Work

Meet customer expectations and business priorities for scheduled field jobs.

Includes

Scheduling Policies

Scheduling & Dispatching Tools

Scheduling Console

1 Completed

1 In Progress

Keep Going

View All

Recent Items

NAME	TYPE	OBJECT
supportforce	Custom App	

30°C Mostly cloudy

Search

ENG

IN

12:34

23-09-2026

Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

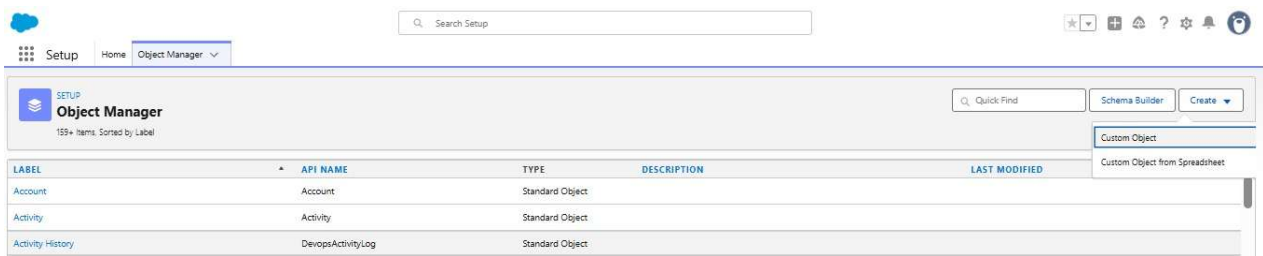
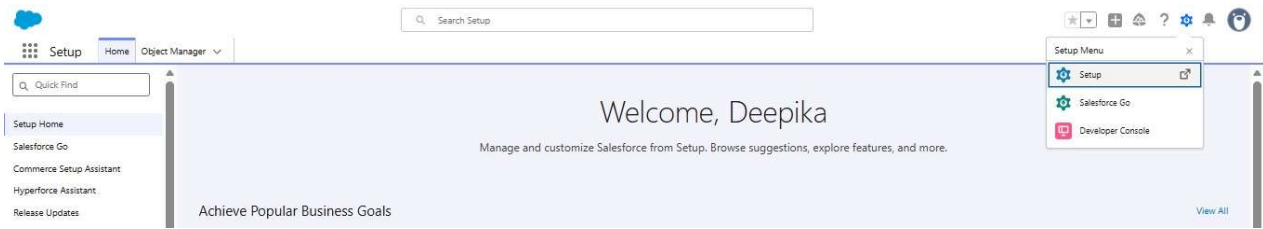
- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

Activity 1 :

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".

4. Fill in the plural label as " Farms".

5. Record name: Farm Name"

6. Select the data type as "Text."

The screenshot shows the 'New Custom Object' setup page in Salesforce. At the top, there's a 'Data Type' dropdown set to 'Text' with a warning message: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.' Below this, the page is divided into several sections: 'Optional Features' with checkboxes for 'Allow Reports' (checked), 'Allow Activities' (unchecked), 'Track Field History' (unchecked), 'Allow in Chatter Groups' (unchecked), and 'Enable Licensing' (unchecked); 'Object Classification' with a note about Enterprise vs. Light Application objects and checkboxes for 'Allow Sharing' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked); 'Deployment Status' with radio buttons for 'In Development' (unchecked) and 'Deployed' (checked); 'Search Status' with a note about searchability and a checked 'Allow Search' checkbox; and 'Object Creation Options (Available only when custom object is first created)' with checkboxes for 'Add Notes and Attachments related list to default page layout' (checked) and 'Launch New Custom Tab Wizard after saving this custom object' (unchecked). At the bottom right, there are buttons for 'Save', 'Save & New', and 'Cancel'.

7. In the Optional Features section, select Allow Reports and Track Field History.

8. In the Deployment Status section, ensure Deployed is selected

.

9. In the Search Status section, select Allow Search.

10. In the Object Creation Options section, select Add Notes and Attachments related list default page layout.

11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
 - i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer.
 - c. Plural Label: Farmers
 - d. Record Name: Farmer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox.
 - g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
 2. Click on the object manager tab just beside the home tab.
 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
-
- a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox.
 - g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Milestone 3:

Tab : Tab

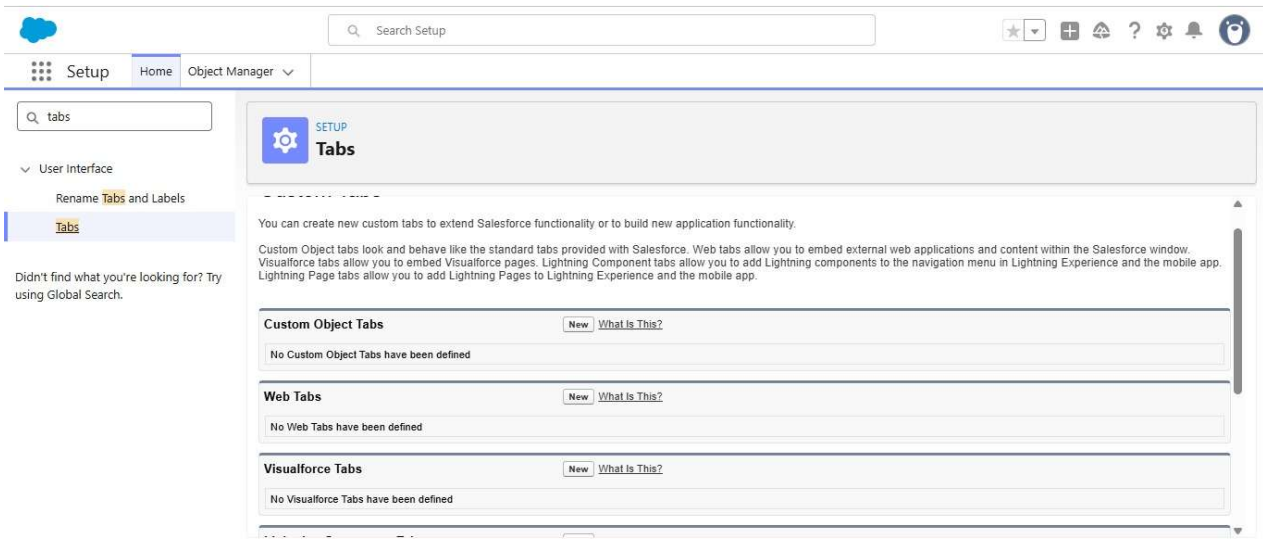
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

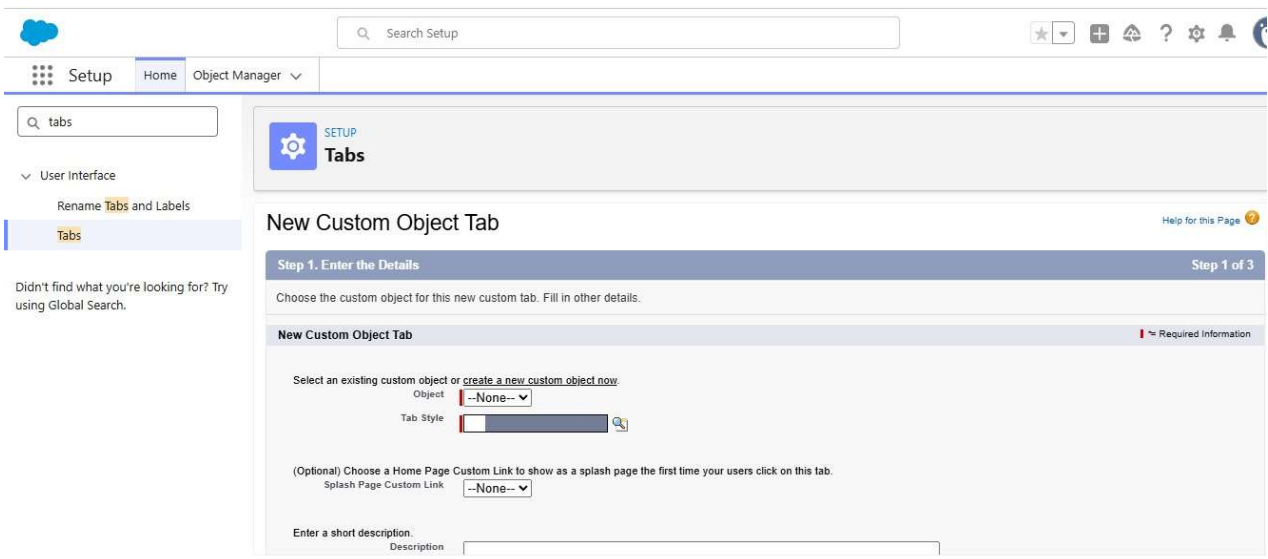
Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. Under custom object tabs, click New.



3. For Object, select Farm.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

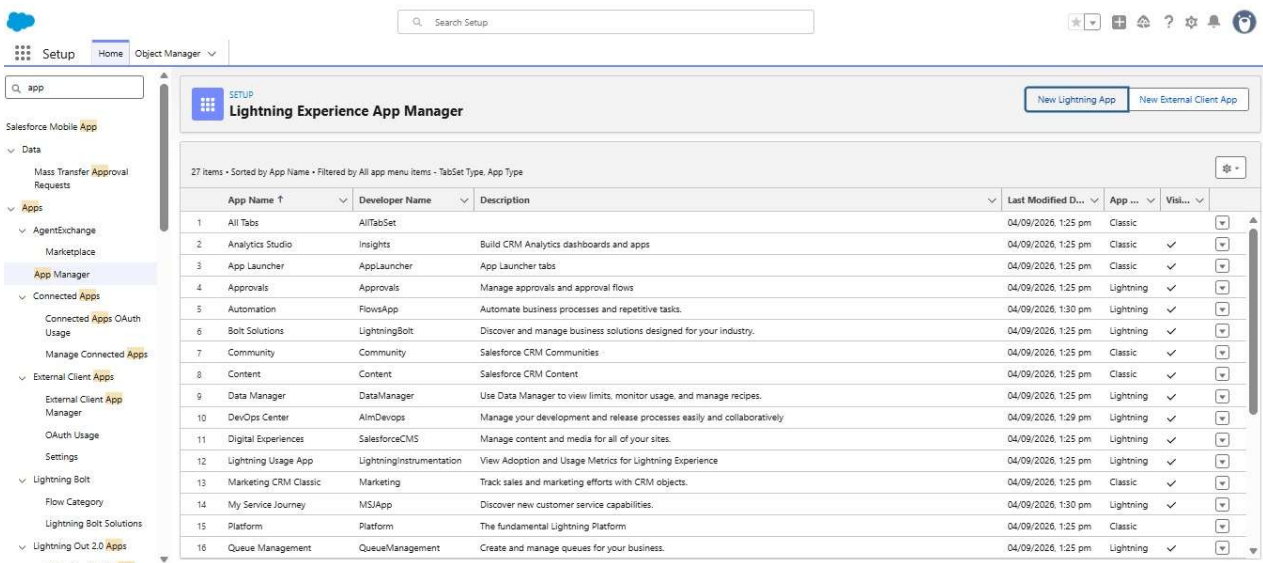
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app



The screenshot shows the Salesforce Lightning Experience App Manager interface. The left sidebar contains a navigation menu with categories like Data, Apps, Connected Apps, External Client Apps, and Lightning Bolt. The main content area displays a table of 27 items, sorted by App Name. The table includes columns for App Name, Developer Name, Description, Last Modified Date, App Type, and Visibility. The apps listed include All Tabs, Analytics Studio, App Launcher, Approvals, Automation, Bolt Solutions, Community, Content, Data Manager, DevOps Center, Digital Experiences, Lightning Usage App, Marketing CRM Classic, My Service Journey, Platform, and Queue Management.

App Name	Developer Name	Description	Last Modified D...	App ...	Visi...
1 All Tabs	AllTabSet		04/09/2026, 1:25 pm	Classic	
2 Analytics Studio	Insights	Build CRM Analytics dashboards and apps	04/09/2026, 1:25 pm	Classic	✓
3 App Launcher	AppLauncher	App Launcher tabs	04/09/2026, 1:25 pm	Classic	✓
4 Approvals	Approvals	Manage approvals and approval flows	04/09/2026, 1:25 pm	Lightning	✓
5 Automation	FlowsApp	Automate business processes and repetitive tasks.	04/09/2026, 1:30 pm	Lightning	✓
6 Bolt Solutions	LightningBolt	Discover and manage business solutions designed for your industry.	04/09/2026, 1:25 pm	Lightning	✓
7 Community	Community	Salesforce CRM Communities	04/09/2026, 1:25 pm	Classic	✓
8 Content	Content	Salesforce CRM Content	04/09/2026, 1:25 pm	Classic	✓
9 Data Manager	DataManager	Use Data Manager to view limits, monitor usage, and manage recipes.	04/09/2026, 1:25 pm	Lightning	✓
10 DevOps Center	AlmDevOps	Manage your development and release processes easily and collaboratively	04/09/2026, 1:29 pm	Lightning	✓
11 Digital Experiences	SalesforceCMS	Manage content and media for all of your sites.	04/09/2026, 1:25 pm	Lightning	✓
12 Lightning Usage App	LightningInstrumentation	View Adoption and Usage Metrics for Lightning Experience	04/09/2026, 1:25 pm	Lightning	✓
13 Marketing CRM Classic	Marketing	Track sales and marketing efforts with CRM objects.	04/09/2026, 1:25 pm	Classic	✓
14 My Service Journey	MSJApp	Discover new customer service capabilities.	04/09/2026, 1:30 pm	Lightning	✓
15 Platform	Platform	The fundamental Lightning Platform	04/09/2026, 1:25 pm	Classic	✓
16 Queue Management	QueueManagement	Create and manage queues for your business.	04/09/2026, 1:25 pm	Lightning	✓

- From Setup, enter App Manager in the Quick Find and select App Manager.
- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next.
- From Available Profiles, select System Administrator and move it to Selected Profiles Profiles Click Save & Finish.
- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

*App Name ⓘ
Name your app...

*Developer Name ⓘ
Enter a developer name...

Description ⓘ
Enter a description...

App Branding

Image ⓘ
Upload

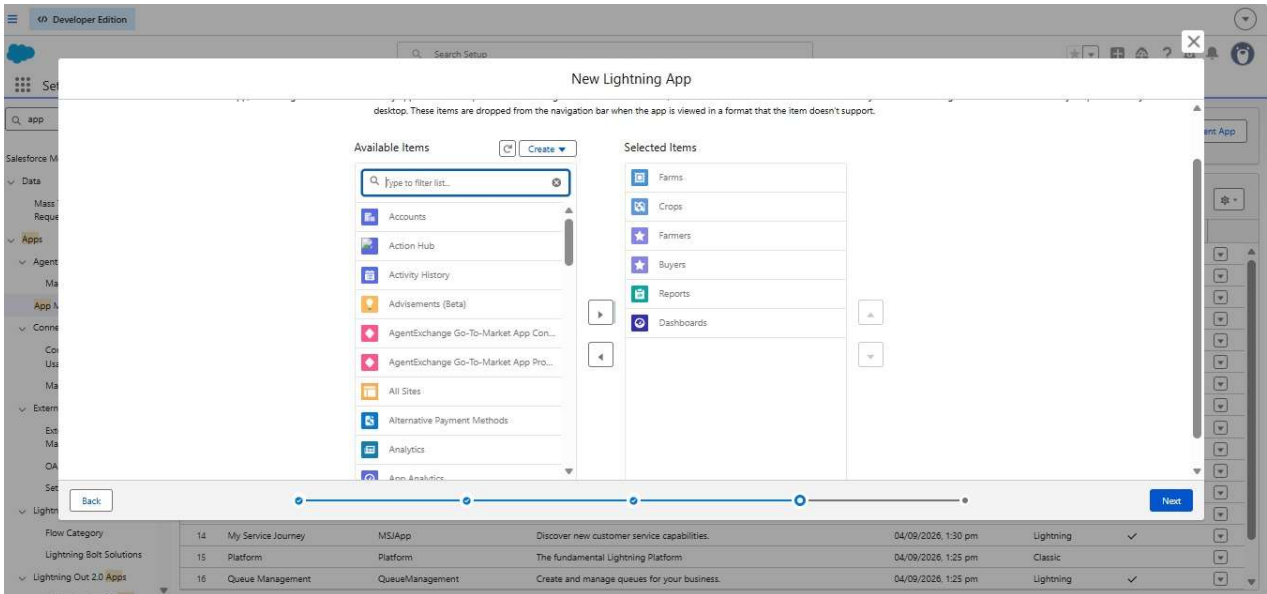
Primary Color Hex
Value ⓘ
#0070D2

Org Theme Options
☐ Use the app's image and color instead of the org's custom theme

App Launcher Preview

Next

3.Navigation menu-Displays the tabs Available inside the app.



Milestone 5:

Fields and Relationship : What are fields?

Fields in salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.

3. Select Farm.

4. Select Fields & Relationships from the left navigation, and click New.

The screenshot shows the Salesforce Setup interface for the 'Farm' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The 'Fields & Relationships' section is active, displaying a table with 4 items, sorted by Field Label. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The items listed are: Created By (CreatedById, Lookup(User)), Farm Name (Name, Text(80)), Last Modified By (LastModifiedById, Lookup(User)), and Owner (OwnerId, Lookup(User.Group)).

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Farm Name	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User.Group)		✓

5 . Choose Data type as a picklist, Click next.

The screenshot shows the Salesforce Setup interface for the 'Farmer' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The 'Fields & Relationships' section is active, displaying a list of data types. The 'Picklist' option is selected, and the 'Next' button is visible at the bottom right.

☐ Checkbox
☐ Currency
☐ Date
☐ Date/Time
☐ Email
☐ Geolocation
☐ Number
☐ Percent
☐ Phone
☒ Picklist
☐ Picklist (Multi-Select)
☐ Text
☐ Text Area
☐ Text Area (Long)
☐ Text Area (Rich)
☐ Text (Encrypted)
☐ Time
☐ URL

Allows users to select a True (checked) or False (unchecked) value.
Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
Allows users to enter a date or pick a date from a popup calendar.
Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the Date/Time field.
Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.
Allows users to define locations. Includes latitude and longitude components, and can be used to calculate distance.
Allows users to enter any number. Leading zeros are removed.
Allows users to enter a percentage number, for example, "10" and automatically adds the percent sign to the number.
Allows users to enter any phone number. Automatically formats it as a phone number.
Allows users to select a value from a list you define.
Allows users to select multiple values from a list you define.
Allows users to enter any combination of letters and numbers.
Allows users to enter up to 255 characters on separate lines.
Allows users to enter up to 131,072 characters on separate lines.
Allows users to enter formatted text, add images and links. Up to 131,072 characters on separate lines.
Allows users to enter any combination of letters and numbers and store them in encrypted form.
Allows users to enter a local time. For example, "2:40 PM", "14:40", and "14:40:00" are all valid times for this field.
Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

Next Cancel

https://islamiahwomenartsandscien-c-dev-ed.develop.my.salesforce-setup.com/one/one.app#/setup/ObjectManager/01ha0000017AIT/view

6. Enter Label(Farm Type), and Enter Values as (Crop Values as (Crop, Livestock,Mixed).

7. Next,Next Click & Save.

The screenshot shows the Salesforce Setup interface for creating a new custom field for the 'Farmer' object. The wizard is at Step 2 of 4, 'Enter the details'. The 'Field Label' is 'Farm Type'. The 'Values' section shows a list of values: 'Crop', 'Livestock', and 'Mixed'. The 'Field Name' is 'Farm_Type'. The 'Description' and 'Help Text' fields are empty.

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)

2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)

3.Choose data type as a text,enter field label(Location)

4.Choose data type as text,enter field label (size).

5.Create a Master Detail relationShip with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

1. Choose Data type As a text, Enter Field Label name (Crop Name), Next save.
2. Select Data type as date , Enter Field label (Planting Date), then save .
3. Again we select data type as a date, Enter field labe (Harvest date), save
4. Next We Create Lookup relationship on crop object realted to Farm Object.
5. We select adata type as currency, Enter field Label (Actual Yield), Next Save it.
6. Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length(20) and save.
7. Now We create Formula Field For That Select data Type As a Formula, Click Next.

The screenshot shows the Salesforce Setup interface. The top navigation bar includes the Salesforce logo, a search bar, and various utility icons. The left sidebar shows the 'Setup' menu with 'Object Manager' selected. The main content area is titled 'Crop' and shows the 'New Custom Field' wizard. The wizard is at 'Step 1: Choose the field type'. Below this, there is a text input field for 'Specify the type of information that the custom field will contain.' The 'Data Type' section is expanded, showing several options: 'None Selected', 'Auto Number', 'Formula' (selected), 'Roll-Up Summary', 'Lookup Relationship', 'Master-Detail Relationship', 'External Lookup Relationship', and 'Checkbox'. Each option has a brief description of its functionality.

Data Type	Description
☐ None Selected	Select one of the data types below.
☐ Auto Number	A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
☒ Formula	A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
☐ Roll-Up Summary	A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
☐ Lookup Relationship	Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
☐ Master-Detail Relationship	Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: - The relationship field is required on all detail records. - The ownership and sharing of a detail record are determined by the master record. - When a user deletes the master record, all detail records are deleted. - You can create roll-up summary fields on the master record to summarize the detail records. The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
☐ External Lookup Relationship	Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.
☐ Checkbox	Allows users to select a True (checked) or False (unchecked) value.

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and Save.

The screenshot shows the Salesforce Setup interface for creating a new custom field for the 'Crop' object. The wizard is at Step 3, 'Enter formula'. The formula entered is 'Expected Yield per Acre (Currency) = Actual_Yield_c / Total_Acre_Harvested_c'. The formula type is set to 'Advanced Formula'. The return type is set to 'Currency'. The wizard includes a 'Functions' list on the right and a 'Quick Tips' section.

Activity 3:

Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its City.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.

The screenshot shows the Salesforce Object Manager interface for the 'Farmer' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'New Field Dependency' and includes instructions on creating a dependent relationship. It shows two steps: Step 1 (Select a controlling field and a dependent field) and Step 2 (Edit filter rules). The current step shows a table with 'Controlling Field' set to 'State' and 'Dependent Field' set to 'District'. Below this, there are 'Continue' and 'Cancel' buttons.

Search Setup

Setup Home Object Manager

SETUP > OBJECT MANAGER
Farmer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

New Field Dependency

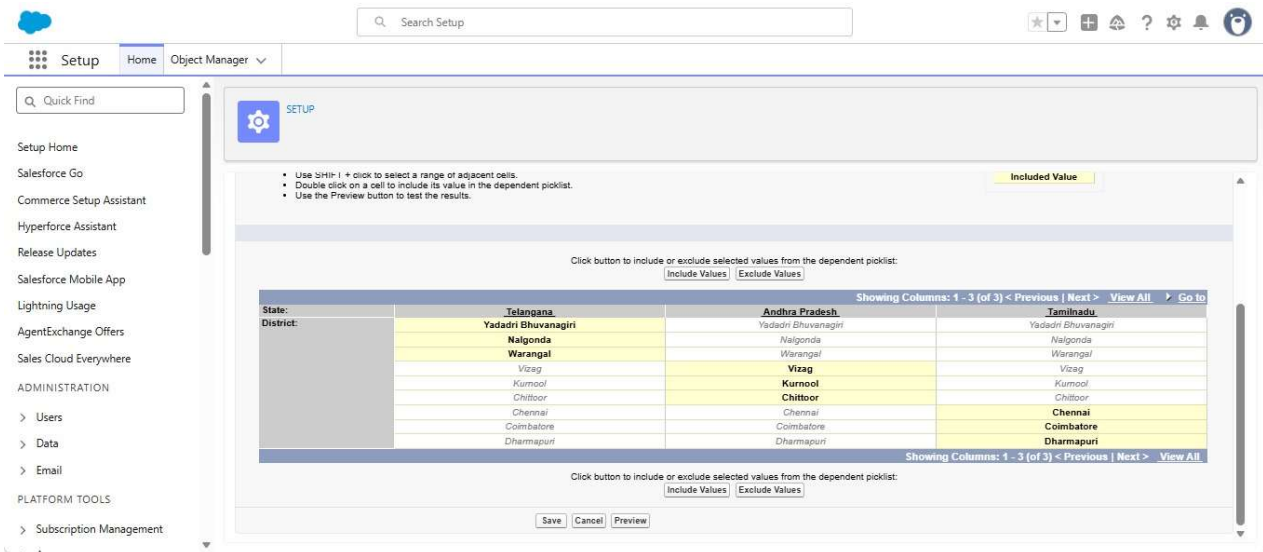
Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.

- The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 300 values can be controlling fields.
- The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.

Step 1: Select a controlling field and a dependent field. Click Continue when finished.

Step 2: On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.

Controlling Field	State	Continue	Cancel
Dependent Field	District	Continue	Cancel



7. Click, Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do

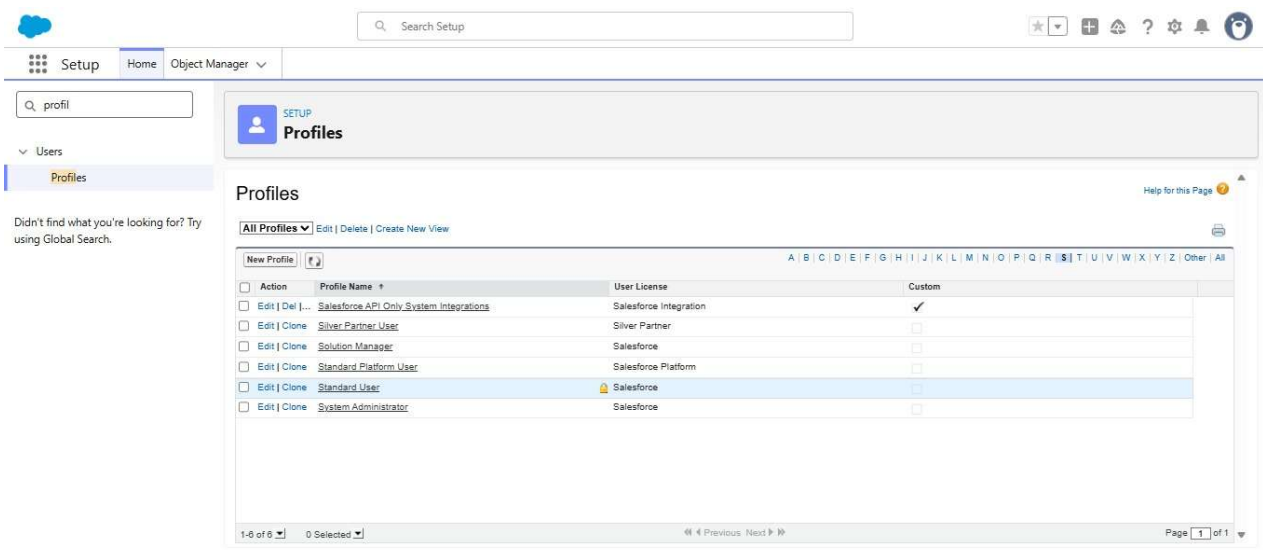
in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

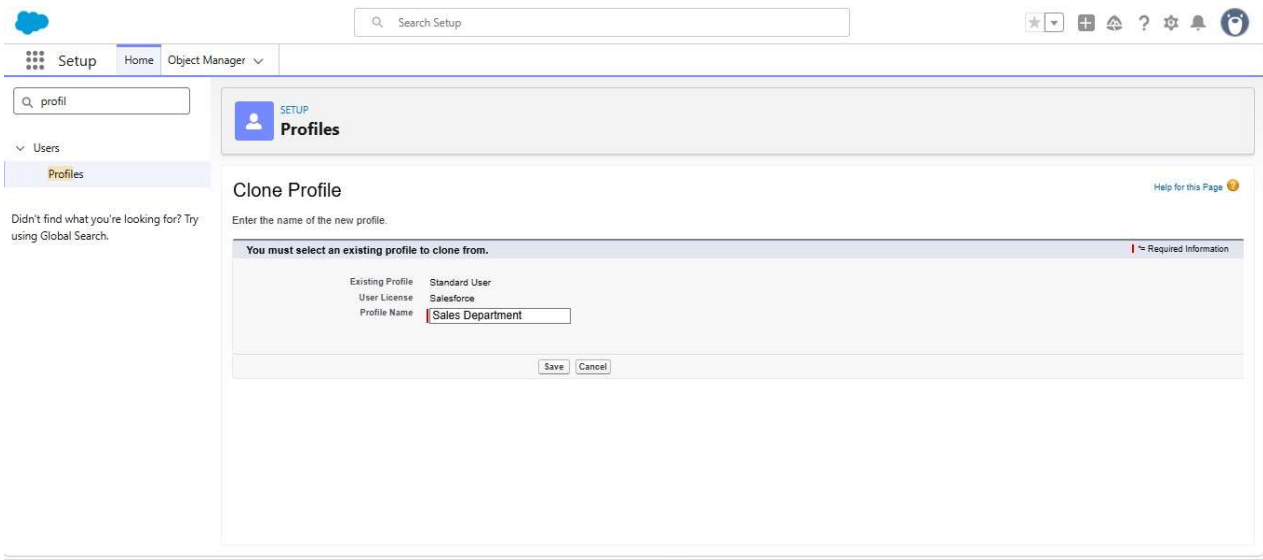
Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, Enter Sales Department.



The screenshot shows the Salesforce Setup interface. The left sidebar has a search bar with 'profil' entered and a list of items including 'Users' and 'Profiles'. The main content area is titled 'Profiles' and shows a table of existing profiles. The 'Standard User' profile is selected and highlighted in blue. The 'User License' column shows 'Salesforce' for the selected profile. The 'Custom' checkbox is checked for the 'Standard User' profile.

Action	Profile Name	User License	Custom
☐ Edit Del ...	Salesforce API Only System Integrations	Salesforce Integration	☒
☐ Edit Clone	Silver Partner User	Silver Partner	☐
☐ Edit Clone	Solution Manager	Salesforce	☐
☐ Edit Clone	Standard Platform User	Salesforce Platform	☐
☐ Edit Clone	Standard User	Salesforce	☒
☐ Edit Clone	System Administrator	Salesforce	☐



5. Click save

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box

7. Select profiles (Standard user).

8. Click clone.

Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and

functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

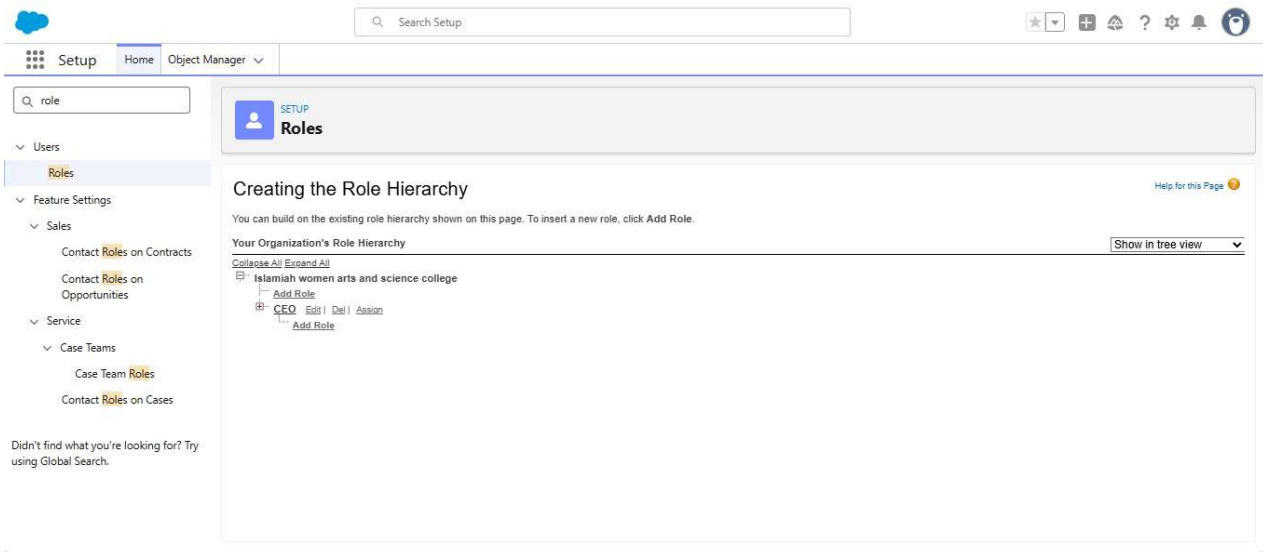
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option.

The screenshot displays the Salesforce Setup interface, specifically the 'Roles' section. The left sidebar shows a navigation menu with 'Users' and 'Roles' highlighted. The main content area is titled 'Understanding Roles' and includes a search bar with 'role' entered. Below the search bar, there's a section for 'Sample Role Hierarchy' with a dropdown menu set to 'Territory-based Sample'. The hierarchy diagram shows a top-level 'Executive Staff' box containing 'CEO, President' and 'CFO, VP, Sales'. Below this are three boxes for 'Western Sales Director', 'Eastern Sales Director', and 'International Sales Director'. Each director box has a list of sales representatives below it: 'Western Sales Rep' (CA, OR), 'Eastern Sales Rep' (NY, MA), and 'International Sales Rep' (Asian, European). To the right of the hierarchy, there are three text boxes explaining permissions: 'View & edit data, roll up forecasts, & generate reports for all users below. Can't access data of other Executive Staff', 'View & edit data, roll up forecasts, & generate reports for all users directly below. Can't access data of users above or at same level', and 'View & edit data, roll up forecasts, & generate reports only for own data. Can't access data of users above or at same level'. At the bottom right, there are buttons for 'Set Up Roles' and 'Don't show this page again'.

3. Below the CEO click on add role and enter the label name as a “Farm Manager” and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license

type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

The screenshot shows the Salesforce Setup interface. In the left sidebar, the 'Users' option is selected under the 'Setup' tab. The main content area is titled 'Active Users' and includes a search bar and a table of active users. The table has columns for Action, Full Name, Alias, Username, Role, Active, and Profile. There are four users listed: DA, Chatter Expert, User Integration, and User Security.

Action	Full Name	Alias	Username	Role	Active	Profile
Edit	A. Deepika	DA	deepikaanath0405@gmail.com	CEO	✓	System Administrator
Edit	Chatter Expert	Chatter	chatty00ahq000005dwmoea0.k7bfpjotw@chatter.salesforce.com		✓	Chatter Free User
Edit	User Integration	integ	integration@00ahq000005dwmoea0.com		✓	Analytics Cloud Integration User
Edit	User Security	asec	insightsecurity@00ahq000005dwmoea0.com		✓	Analytics Cloud Security User

3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write .
- 9) And save the rule.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'sharing' and a list of navigation items: Security, Guest User, Sharing, Rule Access, Report, and Sharing Settings (highlighted). The main content area is titled 'Sharing Settings' and is divided into four steps:

- Step 1: Rule Name**: Fields for Label (Crop), Rule Name (Crop), and Description.
- Step 2: Select your rule type**: Radio buttons for 'Based on record owner' and 'Based on criteria' (selected).
- Step 3: Select which records to be shared**: A table with columns 'Criteria', 'Field', 'Operator', 'Value', and 'AND'. The first row has 'Candidate Name', 'equals', and 'true'. Below the table is a checkbox 'Include records owned by users who can't have an assigned role' which is checked.
- Step 4: Select the users to share with**: (Partially visible at the bottom).

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.

- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows a web application interface for 'Farm Management'. A modal window titled 'Edit Deepika A' is open, displaying a form for editing farmer details. The form includes fields for Farmer Name (Deepika A), Contact Number (1234598760), State (Tamilnadu), District (Chennai), and Village (Chennai). It also shows the Owner (Deepika A), Created By (Deepika A, 25/09/2026, 1:11 pm), and Last Modified By (Deepika A, 25/09/2026, 1:11 pm). The form has 'Cancel', 'Save & New', and 'Save' buttons at the bottom. The background shows a sidebar with 'Farmers' and 'Recently Viewed' sections, and a top navigation bar with 'New', 'Import', 'Change Owner', and 'Assign Label' buttons.

Milestone 11-Reports:

What are Reports?

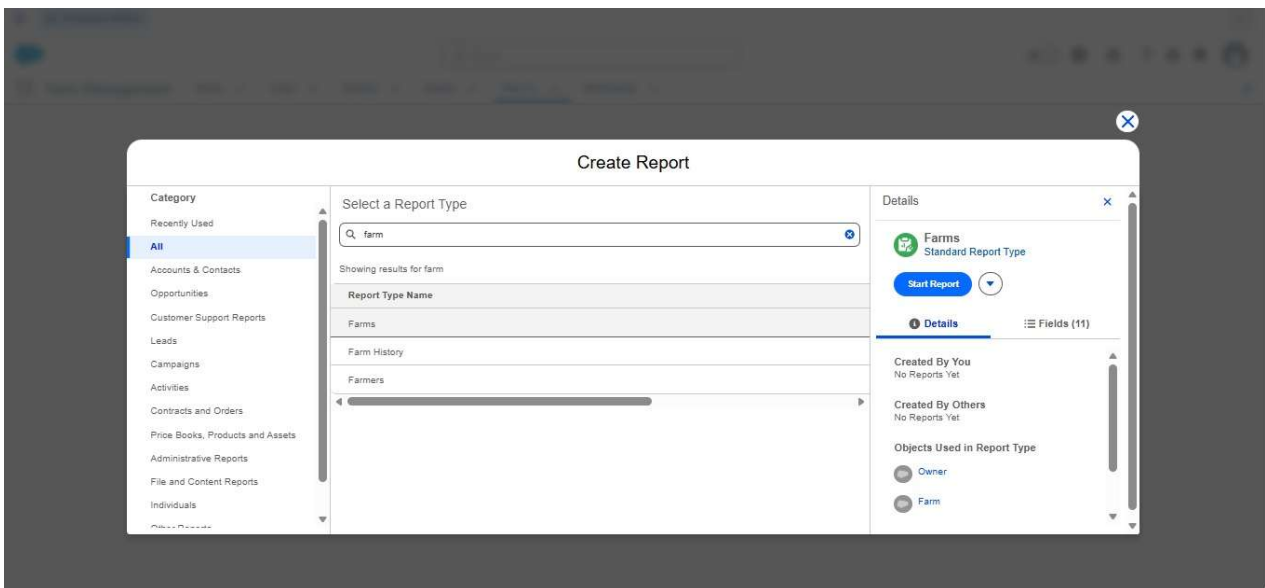
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group, Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.

The screenshot shows the 'FARM MANAGER' application interface. At the top, there's a search bar and navigation tabs for Farms, Crops, Farmers, Buyers, Reports, and Dashboards. The 'Reports' tab is active, displaying a report titled 'Farmers with Farms Report'. The report is a table with columns: Irrigation Type, Farmer: Farmer Name, Farm: Farm Name, Contact Number, Farm Size(Acres), and Farm Type. The data is grouped by 'Irrigation Type' with a sub-total for each group. The 'Total (3)' row shows a grand total of 25 for Farm Size(Acres).

Irrigation Type	Farmer: Farmer Name	Farm: Farm Name	Contact Number	Farm Size(Acres)	Farm Type
Localized Irrigation (3)	Raghu	Paddy Farm	988430778	10	Livestock
	Raghu	Rural Farm	988430778	10	Crop
	Raghu	Wheat Farm	788262779	15	Livestock
Subtotal				25	
Total (3)				25	

At the bottom of the report, there are toggle switches for 'Row Counts', 'Detail Rows', 'Subtotals', and 'Grand Total', all of which are currently turned on. There is also a 'Conditional Formatting' link on the right.

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

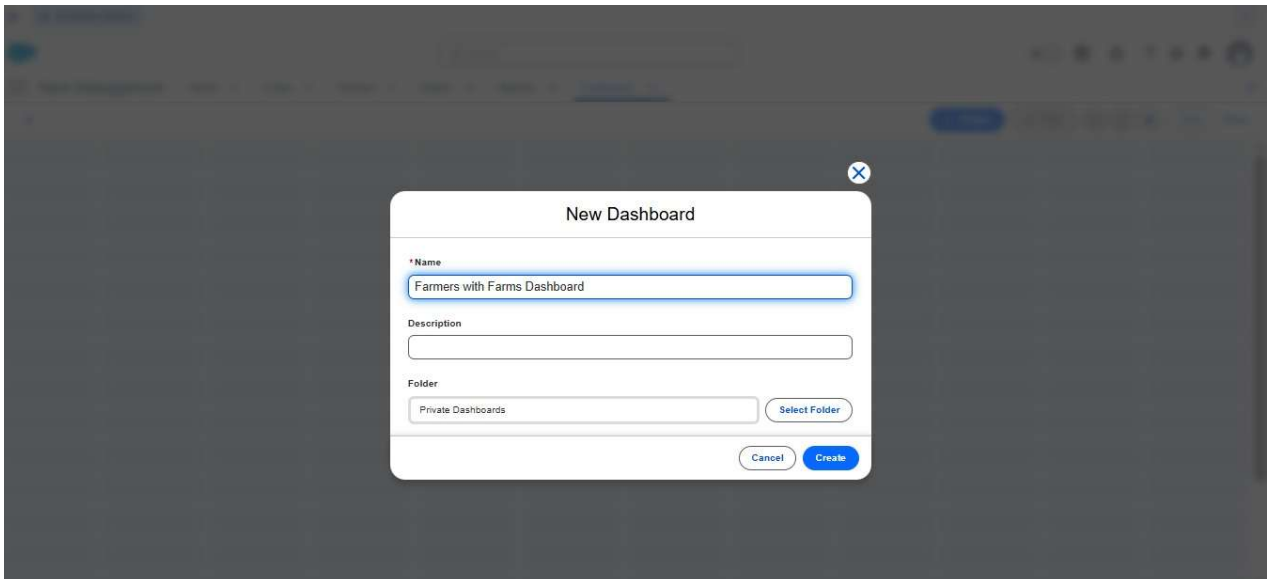
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.

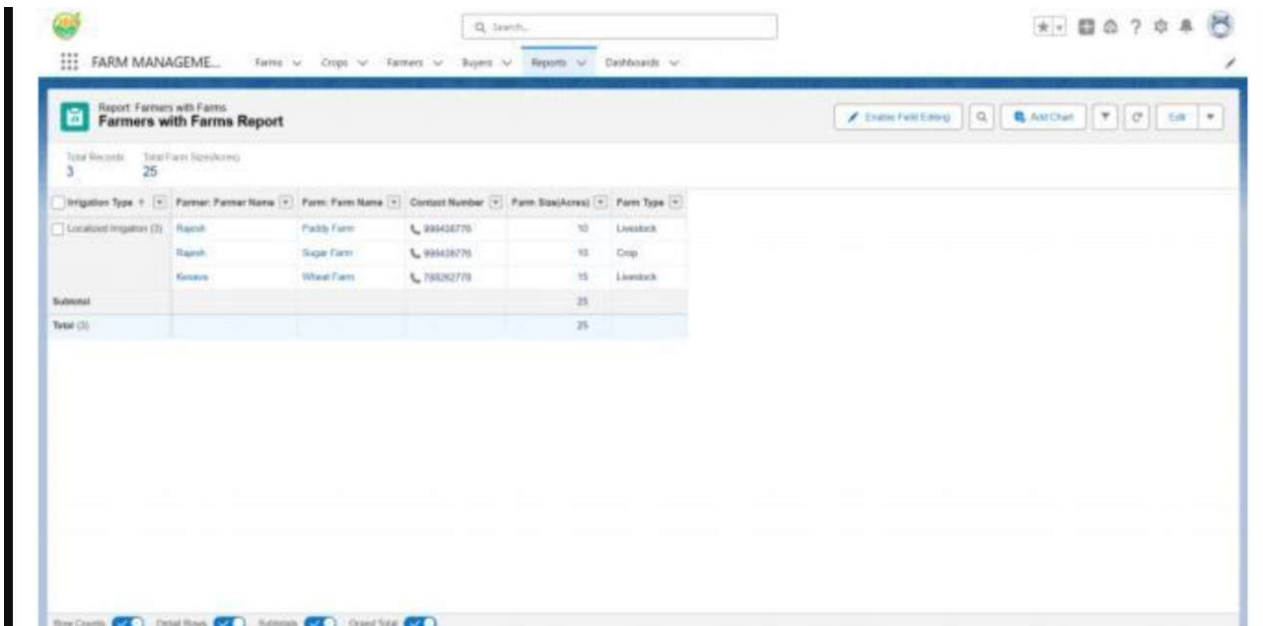


Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.

- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

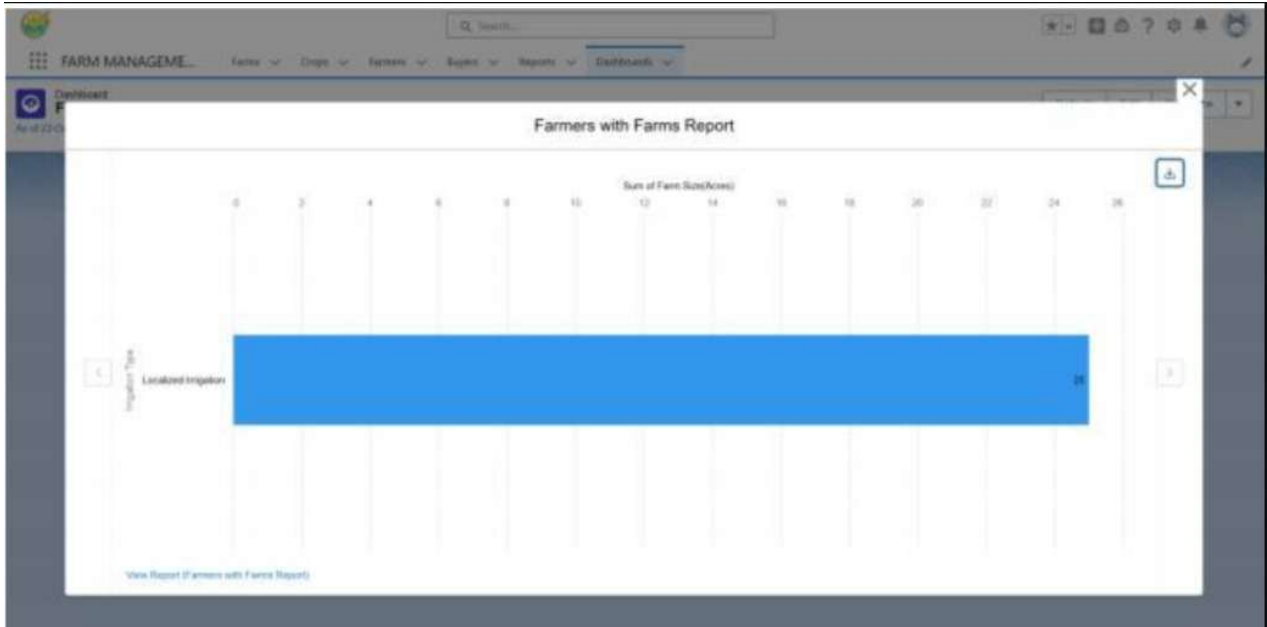


The screenshot shows a web application interface for 'FARM MANAGEMENT'. The top navigation bar includes tabs for 'Farms', 'Crops', 'Farmers', 'Supers', 'Reports', and 'Dashboards'. The 'Reports' tab is selected, and the sub-report 'Farmers with Farms Report' is displayed. The report shows a table with columns: 'Irrigation Type', 'Farmer: Farmer Name', 'Farm: Farm Name', 'Contact Number', 'Farm Size(Acres)', and 'Farm Type'. The table contains three rows of data and a 'Total (3)' row. The 'Total Farm Size(Acres)' is 25. The 'Irrigation Type' is 'Localized Irrigation (3)'. The 'Farm Type' is 'Livestock'.

Irrigation Type	Farmer: Farmer Name	Farm: Farm Name	Contact Number	Farm Size(Acres)	Farm Type
Localized Irrigation (3)	Rajesh	Paddy Farm	999436779	10	Livestock
	Rajesh	Sugar Farm	999436779	10	Crop
	Kavasa	Wheat Farm	795262779	15	Livestock
Subtotal				25	
Total (3)				25	

Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user Operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration

- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org.

Thank you

